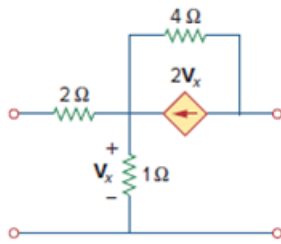


Problem

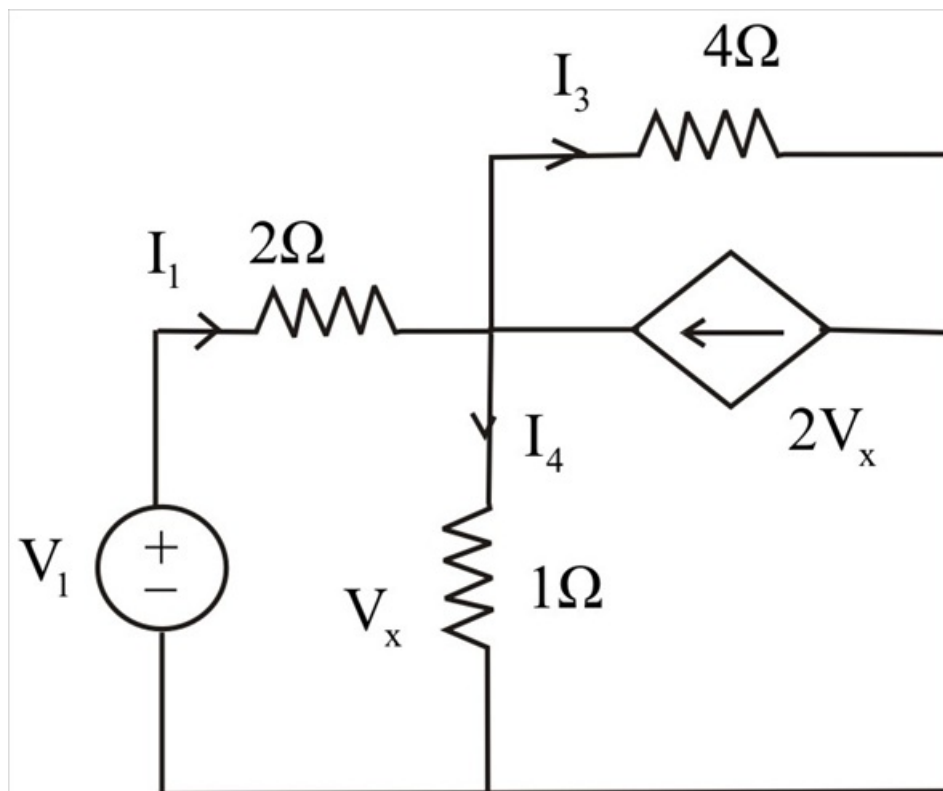
Calculate $[y]$ for the two-port in Fig. 19.85.

Figure 19.85



Step-by-step solution

Step 1 of 7



Step 2 of 7

For $V_2 = 0$ we have the circuit as above :-

At Node 1

$$I_1 = \frac{V_x}{1} + \frac{V_x}{4} - 2V_x$$

$$I_1 = \frac{4V_x + V_x - 8V_x}{4} = \frac{11V_x}{4}$$

$$V_x = V_1 - I_1 \cdot 2$$

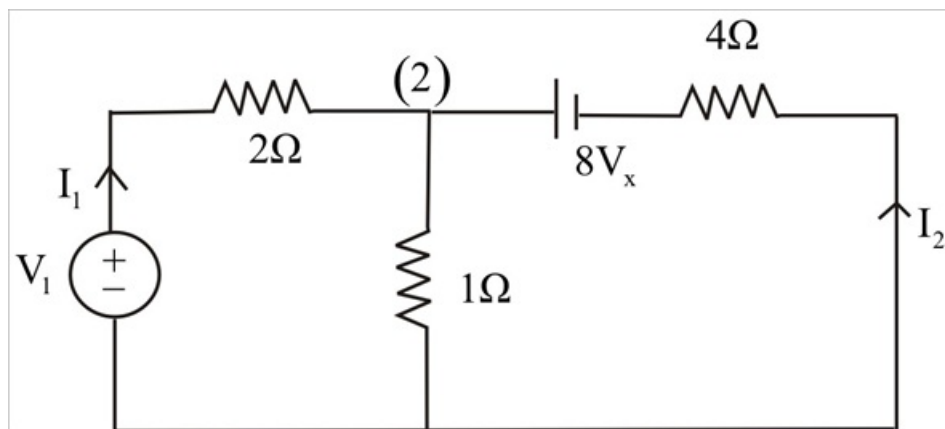
$$I_1 = \frac{11(V_1 - 2I_1)}{4} \Rightarrow \frac{11V_1}{4} = I_1 + \frac{I_1}{2}$$

$$\left. \frac{I_1}{V_1} \right|_{V_2=0} = \frac{1}{11} = \frac{11}{6} S.$$

At Node 2

$$I_1 = \frac{V_x}{1} - I_2$$

Step 3 of 7



Step 4 of 7

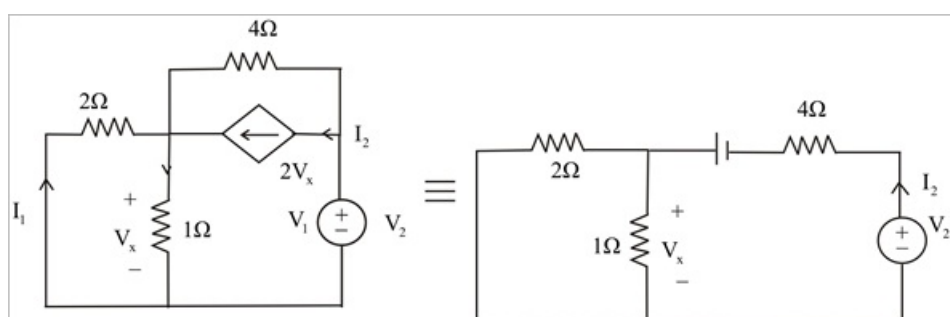
$$\frac{V_1 - V_x}{2} = \frac{V_x}{1} - I_2$$

$$\frac{11}{6}V_1 = \frac{V_1 - 2 \times \frac{11}{6}V_1}{1} - I_2$$

$$I_2 = V_1 \left(\frac{6 - 22}{6} - \frac{11}{6} \right) = \left. \frac{I_2}{V_1} \right|_{V_2=0} = \frac{-27}{6} = \frac{-9}{2} S$$

for $V_1 = 0$ we have

Step 5 of 7



The above circuit reduces to

$$\text{At Node 2} \quad I_1 = \frac{V_x}{1} - I_2$$

$$V_x = V_2 + 8V_x - 4I_2$$

$$V_x = \frac{4I_2 - V_2}{7}$$

$$\frac{4I_2 - V_2}{7} = I_2$$

$$\Rightarrow \frac{-V_2}{14} = I_2 \frac{-2}{7} I_2 \Rightarrow \frac{I_2}{V_2} = \frac{7}{19 \times 5} = \frac{1}{10} S$$

$$I_1 = V_x - I_2 \Rightarrow I_1 = \frac{4I_2 - V_2}{7} - I_2$$

$$I_1 = \frac{-V_2}{7} - \frac{3I_2}{7} \Rightarrow I_1 = \frac{-V_2}{7} - \frac{3}{70} V_2$$

$$I_1 = \frac{-13}{70} V_2 \Rightarrow \frac{I_1}{V_2} = \frac{-13}{70}$$

$$Y = \begin{bmatrix} \frac{11}{6} & \frac{-13}{70} \\ \frac{-9}{2} & \frac{1}{10} \end{bmatrix} S$$